

Uniform transport of arbitrary representation data

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Abstract

The transport half of the seventh benchmark rung, completed in one uniform package. A representation datum of a level-one eigenform seed is a finite direct sum of symmetric powers $\rho = \bigoplus_i \text{Sym}^{r_i}$ — the general finite algebraic representation of the dual group at level one, determinant twists being trivial at det-one normalization. For *every* such ρ simultaneously: the transported coefficient bank is the Dirichlet product of the constituent Sym^r banks; the datum is a polynomial Satake pair with one explicit exponent; its L -series converges absolutely beyond one explicit polynomial threshold with no temperedness input; the L -series factors as the product of the constituent L -series on the common half-plane; and transport of a direct sum is the product of the transports. No per-representation argument occurs anywhere — one induction over the transport-closure calculus carries all of it. Everything is unconditional and machine-checked.

1 The datum and the transport

Fix a level-one holomorphic Hecke eigenform seed with Satake parameters α_p , and let $\rho = (r_1, \dots, r_m)$ be a finite list of symmetric-power ranks. The *channel type* of ρ is built by recursion — one block $\{0, \dots, r_i\}$ per constituent — and the *weight system* of ρ assigns to each prime the disjoint union of the constituent clocks $(\alpha_p^{r_i}, \alpha_p^{r_i-2}, \dots, \alpha_p^{-r_i})$ (**repIndex**, **repWeight**). Because the recursion makes each step *definitionally* a direct sum, the closure calculus applies with no reindexing.

Theorem 1 (uniform bank law). *For every ρ , the transported bank of the datum is the Dirichlet convolution of the constituent banks: $c_\rho = c_{\text{Sym}^{r_1}} \star \dots \star c_{\text{Sym}^{r_m}}$. (**repBank_eq**: one induction; the empty datum carries the convolution identity δ , and each constituent step is the closure law with the rank-uniform clock-bank identification.)*

Theorem 2 (uniform admissibility and summability). *Every datum is a polynomial Satake pair with the single exponent $E \cdot \sum_i r_i$ (**repSatakePair**, bounds inherited from the constituent pairs), and its L -series converges absolutely for $\Re s > B_\rho + 1$ with $B_\rho = \# \text{channels}(\rho) + E \cdot \sum_i r_i$ explicit (**repBank_LSeriesSummable**) — with no temperedness input, uniformly over all representation data.*

Theorem 3 (uniform L -factorization). *On the common half-plane, $L(\rho, s) = \prod_i L(\text{Sym}^{r_i}, s)$. (**LSeries_repBank**, **prodBank_summable**.)*

*Machine-checked companion: `RequestProject/UniformTransport.lean`, 8 named declarations, each at axiom footprint `{propxert, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

Theorem 4 (composition is automatic). $c_{\rho_1 \oplus \rho_2} = c_{\rho_1} \star c_{\rho_2}$ for all data ρ_1, ρ_2 — the closure rung’s law at the level of arbitrary representation data. (**repBank_append**.)

Theorem 5 (rung 7, one theorem). For every representation datum ρ , the conclusions of Theorems 1–4 hold simultaneously. (**uniform_transport**.)

Remark 6 (register, scope, falsifiers). Proven and machine-checked, unconditionally: everything above, with the datum class the finite direct sums of symmetric powers of the $GL(2)$ seed — the finite algebraic representations of its dual group at level one; local factors and per-place reciprocity for arbitrary finite duality-stable weight multisets are the compiled **FiniteWeightFiber** layer of the main paper. The reflection at every rank is the carrier’s own, with the global identification proven at all places (**prop:localid** of the main paper); per-rank registration of that reflection under its classical name in the prescribed chart is the corroboration currency of the companion notes (five instances landed). A counterexample would be a datum whose bank differs from the convolution of its constituent banks at some prime power, or violates its summability threshold, or whose L -series escapes the factorization on the common half-plane — each formally stated, so a counterexample would require a kernel failure.

References

- [1] R. P. Langlands, *Problems in the theory of automorphic forms*, in: Lectures in Modern Analysis and Applications III, Lecture Notes in Math. **170**, Springer (1970), 18–61.